Gram molecular mass of H_2O

$= (1 \times 2) + (16 \times 1)$

$= 2 + 16$

Gram molecular mass of $H_2O = 18$ g

2) CO_2

Atomic masses of C = 12, O = 16

Gram molecular mass of CO_2

$= (12 \times 1) + (16 \times 2)$

$= 12 + 32$

Gram molecular mass of $CO_2 = 44$ g

3) $Ca_3(PO_4)_2$

Atomic masses of Ca = 40, P = 30, O = 16.

Gram molecular mass of $Ca_3(PO_4)_2$

$= (40 \times 3) + [30 + (16 \times 4)] \times 2$

$= 120 + (94 \times 2)$

$= 120 + 188$

Gram molecular mass of $Ca_3(PO_4)_2 = 308$ g

II. Calculation based on number of moles from mass and volume

1) Calculate the number of moles in 46 g of sodium?

$$\text{Number of moles} = \frac{\text{Mass of the element}}{\text{Atomic mass of the element}}$$

$= 46 / 23$

= 2 moles of sodium

2) 5.6 litre of oxygen at S.T.P

$$\text{Number moles} = \frac{\text{Given volume of } O_2 \text{ at S.T.P}}{\text{Molar volume at S.T.P}}$$

$$\text{Number of moles of oxygen} = \frac{5.6}{22.4}$$

= 0.25 mole of oxygen

3) Calculate the number of moles of a sample that contains 12.046×10^{23} atoms of iron ?

$$\text{Number of moles} = \frac{\text{Number of atoms of iron}}{\text{Avogadro's number}}$$

$= 12.046 \times 10^{23} / 6.023 \times 10^{23}$

= 2 moles of iron

III. Calculation of mass from mole

Calculate the mass of the following

1) 0.3 mole of aluminium (Atomic mass of Al = 27)

$$\text{Number of moles} = \frac{\text{Mass of Al}}{\text{Atomic mass of Al}}$$

Mass = No. of moles × atomic mass

So, mass of Al = 0.3×27

= 8.1 g

2) 2.24 litre of SO_2 gas at S.T.P

Molecular mass of $SO_2 = 32 + (16 \times 2)$

$= 32 + 32 = 64$

$$\text{Number of moles of } SO_2 = \frac{\text{Given volume of } SO_2 \text{ at S.T.P}}{\text{Molar volume } SO_2 \text{ at S.T.P}}$$

$$\text{Number of moles of } SO_2 = \frac{2.24}{22.4}$$

= 0.1 mole

$$\text{Number of moles} = \frac{\text{Mass}}{\text{Molecular mass}}$$

Mass = No. of moles × molecular mass

Mass = 0.1×64

Mass of $SO_2 = 6.4$ g

3) 1.51×10^{23} molecules of water

Molecular mass of $H_2O = 18$

$$\text{Number of moles} = \frac{\text{Number of molecules of water}}{\text{Avogadro's number}}$$

$= 1.51 \times 10^{23} / 6.023 \times 10^{23}$

= 1 / 4

= 0.25 mole

$$\text{Number of moles} = \frac{\text{Mass}}{\text{Molecular mass}}$$

0.25 = mass / 18

Mass = 0.25×18

Mass = 4.5 g

4) 5×10^{23} molecules of glucose ?

Molecular mass of glucose = 180

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